



Roll Number

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2025
Odd Semester - 2025-26
HU101 - ENVIRONMENTAL SCIENCE

Time: 3 Hrs

Maximum Marks : 60

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Part A

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. Which is not an objective of EIA ? (1) C02 BL2 PO2
(a) Recycling and reduction of waste (b) Risk analysis and disaster management
(c) Assessment of international funding (d) All of the above
2. Which of the following is a man-made disaster? (1) C01 BL2 PO2
(a) Tsunami (b) Volcanic eruption
(c) Landslide (d) Oil spill
3. Energy flow in an eco-system is- (1) C01 BL1 PO1
(a) unidirectional (b) cyclic
(c) may be unidirectional or cyclic depends on condition (d) can't be said
4. Biomedical waste is an example of- (1) C02 BL2 PO3
(a) Hazardous waste (b) Biodegradable waste
(c) Non-biodegradable waste (d) None of these
5. Vehicular pollution can be controlled by- (1) C02 BL2 PO3
(a) ESP (b) Catalytic converter
(c) Bag house (d) All of these
6. Blackfoot disease is related to- (1) C01 BL2 PO1

- (a) As (b) Hg
(c) Pb (d) Cd
7. Which one is the intensity level of noise? (1) C01 BL2 PO2
(a) Centimetre (b) Decibel
(c) Newton (d) Watt
8. BOD test in the laboratory is done for- (1) C04 BL3 PO4
(a) 5 days (b) 6 days
(c) 10 days (d) 2 days
9. The best method of disposal of non-hazardous solid waste- (1) C01 BL3 PO4
(a) Open dumping (b) Sanitary land filling
(c) Incineration (d) Composting
10. Montreal protocol is related with- (1) C01 BL3 PO1
(a) Ozone layer depletion (b) Greenhouse gas emission
(c) SPM (d) Water pollution
11. The coldest region of the atmosphere is (1) C02 BL1 PO1
(a) Troposphere (b) Stratosphere
(c) Mesosphere (d) Thermosphere
12. Organomercury is an example of (1) C01 BL1 PO1
(a) Fungicide (b) Fumigant
(c) Antibiotic (d) Rodenticide

Part B**(Answer any four of the following) 5x4=20**

13. Briefly describe Exponential growth of population and establish, $N_t = N_0 e^{RT}$ (5) C03 BL2 PO3
14. Briefly describe the four major steps involved in earthquake disaster management. (5) C02 BL2 PO7
15. If two machines produce 50 dB sounds simultaneously, what will be the total sound level? (5) C03 BL4 PO6
16. Define eutrophication and state the environmental effects caused by eutrophication. (5) C02 BL3 PO3

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| 17. What is an ecosystem? Briefly describe the components of an ideal ecosystem. | (5) | C01 | BL2 | PO1 |
| 18. What is Maximum sustainable Yield? Prove that Maximum sustainable Yield is achieved when $N=K/2$, where the notations have their own meaning., | (5) | C01 | BL3 | PO1 |

Part C**(Answer any three of the following) 3x10=30**

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|---|------|-----|-----|-----|
| 19. a) Derive $BOD_t = C_0 (1 - e^{-kt})$.

b) A waste water sample of 25 ml is diluted by adding 275 ml seeded dilution water. DO_i is found to be 9 mg/l and after five days the final DO_f is 3 mg/l. If seeded dilution water has initial Bi of 10 mg/l and B_f 8 mg/l, find the BOD_5 of the waste. | (10) | C04 | BL4 | PO1 |
| 20. Answer the following- | | | | |
| (a) Noise in an area is measured as 95 dB for 1 hour, 90 dB for 3 hours, 100 dB for 30 minutes and 75 dB for 3 hours. Followed the noise threshold limit, find the working place is good for a worker or not. | (6) | C04 | BL4 | PO6 |
| (b) Briefly describe the Biomedical waste management. | (4) | C02 | BL1 | PO3 |
| 21. (a) How does the acid rain form? Give detail reactions | (5) | C04 | BL2 | PO1 |
| (b) Why the ground water is found to be alkaline? How the alkalinity can be determined? | (5) | C04 | BL2 | PO1 |
| 22. (a) Explain bio-magnifications. What is its significance in food chain? | (5) | C01 | BL1 | PO1 |
| (b) How the Chloride ion can be determined for a given sample water? | (5) | C04 | BL2 | PO1 |
| 23. Answer the following questions: | | | | |
| (a) Differentiate between London Smog and Photochemical Smog. Give reactions for the formation of PAN. | (5) | C02 | BL4 | PO1 |
| (b) Green House effect is useful for the survival of life on earth. Justify. What are impacts of the Green House effect? | (5) | C04 | BL3 | PO1 |